Pythagorean Triples

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1 What is a Pythagorean triple?

A Pythagorean triple is a set of three positive integers (a, b, c) that satisfy the equation of the Pythagorean theorem: $a^2 + b^2 = c^2$. These triples represent the side lengths of a right-angled triangle.

Some example of Pythagorean triples are the following:

- (3, 4, 5)
- (5, 12, 13)
- (20, 21, 29)

2 Method

For our research, our main objective is to find and prove a general rule to find the Pythagorean Triples with the same value of c. Example shown in 1

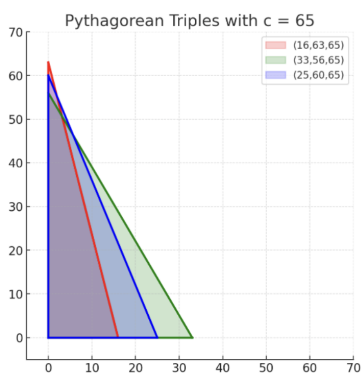

Figure 1: A geometric representation of three distinct Pythagorean triples that all share the same hypotenuse, c=65.

The Pythagorean Triple Rule

The Pythagorean triple rule is based on the **sum of two squares** and the factorization of c in the Gaussian integers (e.g., $a + bi$), derived from complex numbers.

Looking at only the part: sum of two squares

First Point: Every prime number congruent to $1 \pmod 4$ (e.g., $1, 5, 9, 13, 17, 21, 25, \ldots$) can be written as

$$a^2 + b^2, \quad \text{where } a, b \geq 0 \text{ and } a \geq b.$$

Examples:

$$\begin{aligned} 5 &= 1^2 + 2^2 \qquad [a = 2,\, b = 1] \\ 13 &= 2^2 + 3^2 \qquad [a = 3,\, b = 2] \\ 17 &= 1^2 + 4^2 \qquad [a = 4,\, b = 1] \end{aligned}$$

Second Point: This describes how many possible ways there are to write a number C (which is a product of multiple prime numbers, each of which is also $1 \pmod 4$) as a sum of two squares.

$$\text{Number of ways} = 2^{r-1}$$

where r is the number of prime factors that are $1 \pmod 4$.

Examples:

$$\begin{aligned} 5 &= 1^2 + 2^2 && \Rightarrow \text{No. of ways } = 2^{1-1} = 1 \\ 65 &= 1^2 + 8^2 = 4^2 + 7^2 && \Rightarrow \text{No. of ways } = 2^{2-1} = 2 \\ 1105 &= 4^2 + 33^2 = 9^2 + 32^2 = 12^2 + 31^2 = 23^2 + 24^2 && \Rightarrow \text{No. of ways } = 2^{3-1} = 4 \end{aligned}$$

Here, $5, 65, 1105$ are all numbers congruent to $1 \pmod 4$.

Third and Fourth Points:

We can represent a prime number as a product of complex Gaussian integers, $z = a + bi$.

Example:

$$\text{Prime numbers: } z_1 z_1^* \, z_2 z_2^* \, z_3 z_3^* \ldots z_r z_r^*$$

where each z_i and z_i^* form a conjugate pair and represent one prime number.

For the representation of a prime number as a product of Gaussian primes, we have 2 options for each prime number. So, the total number of choices is 2^r.

However, since z and z^* are conjugates, $(a + bi)(a - bi) = a^2 + b^2$, and each pair of conjugates counts only once. Therefore, the number of distinct ways is:

$$\text{No. of ways} = 2^{r-1}$$

Remark: This is just a method to derive the formula using complex numbers.